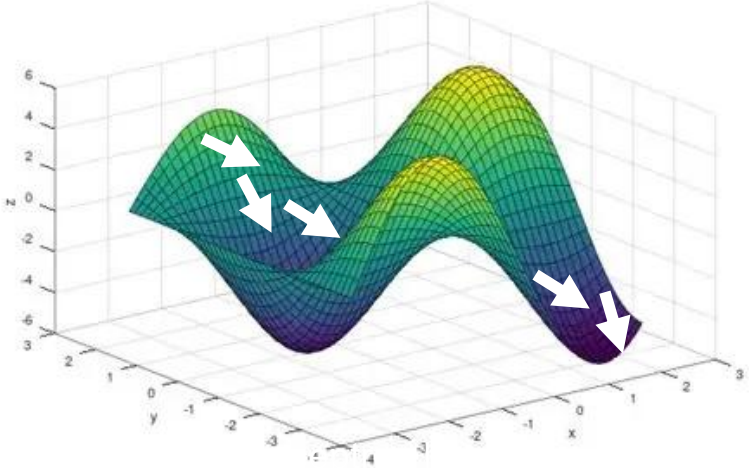
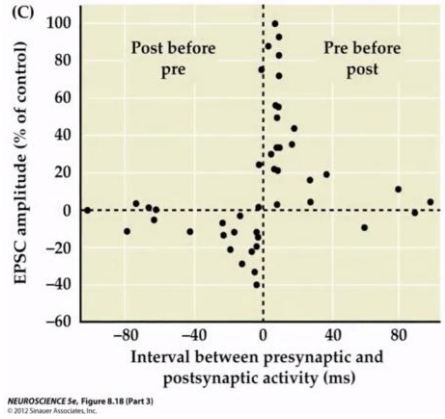
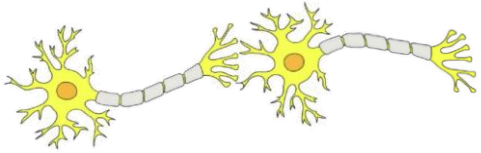
Recap I

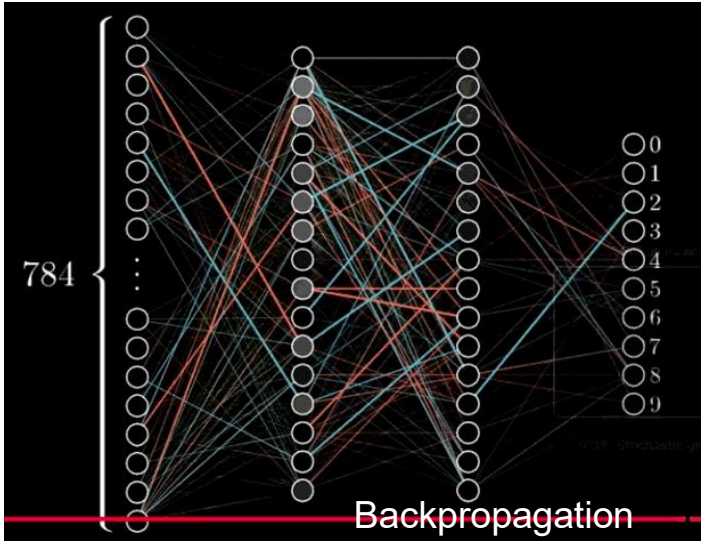


Loss function and gradient descent

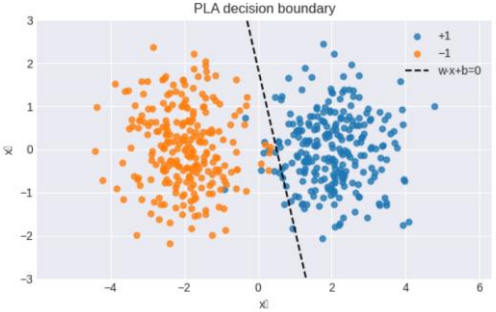


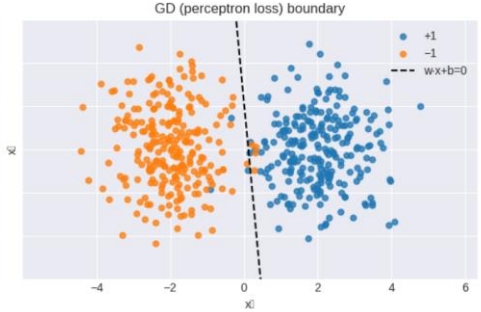
LT depression or
LT potentiation





Recap II






```
PLA time: 0.88
GD time : 0.02
```

Session 5: Developping Brains vs training ANNs

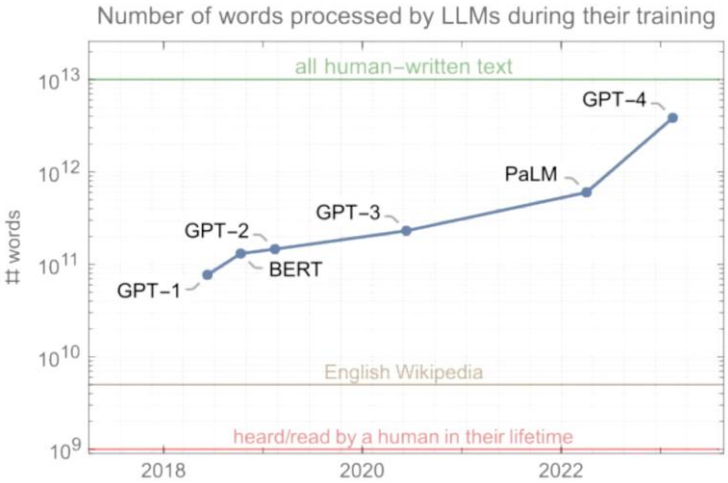
- How do the
  - loss function
  - learning rate
  - gradient descent
  - backpropagation

  - bias & variance
  - regularization

- relate to brain development?

Why compare them?

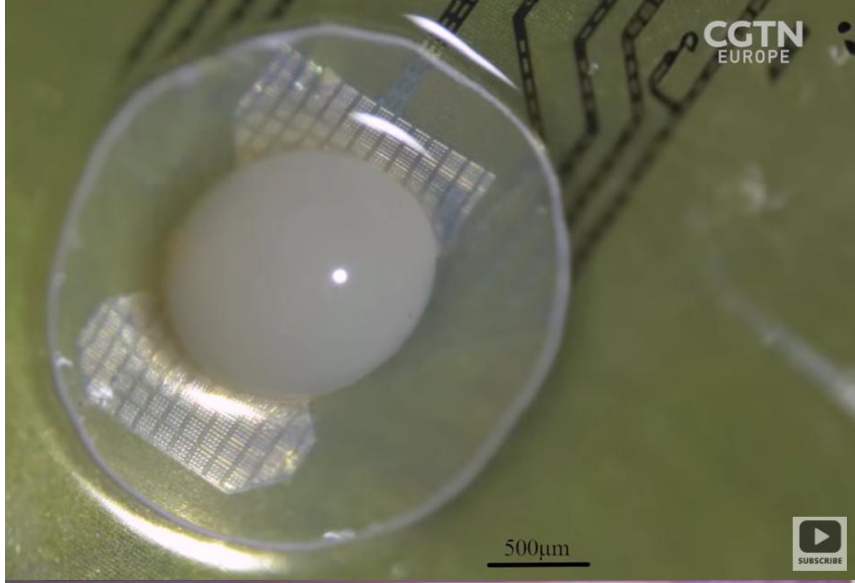
Training ANN is less efficient when compared to human brain



But ANN excell in speed in other tasks: image generation, text summarizing

Training the brain: organoids

- Stem cells are promted to become neurons
- Neurons aggregate to form these organoids
- Attached to electrodes to ineract with them
- Rewarded with dopamine
- Store bits of information and logical operators (AND, OR)
- 120 different types of neurons
- Challenges: vascularization, short life span and ethics



[Biocomputers made from human brain cells could run the AI systems of the future]

Questions

- Can we develop different algorithms for learning in organoids? (preceptron rule 1975, backpropagation 1962, xxxx 2027?)

- Can organoids be upscaled?
  - compostable, efficient...

- How can ANN concepts:
  - loss function,
  - gradient descent,
  - backpropagation,
  - learning rate,
  - regularization,
  - bias vs variance,
  - weight initialization...

be mapped to human processes?

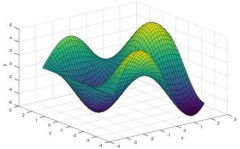
The connectionist approach: pros and cons

1. Biolog. and ANN share general **structural** and **functional** characters: nodes-soma, link-axon, activation function-firing mechanisms, parallel processing
2. Learning via the adjustment of the **weights** - vs **LTP**
3. React to damage with **graceful degradation:** gradual decrease in performance with increased damage.
4. **Generalization**, pattern completion, hallucinations, superstitions
5. **Interference**: two sets of similar info interfere with each other (study two languages)

1. Biological neurons are way more **complex**
2. ANN **estabilize** after converging to a solution, whereas biol. neurons are oscilatory
3. **Catastrophic forgetting**: an ANN has learned to recognize a set of patterns and then is called on to learn a new set.
4. No **backpropagation** in biological networks.

Friedenberg, Cognitive Science: An Introduction to the Study of Mind p 201

Loss function

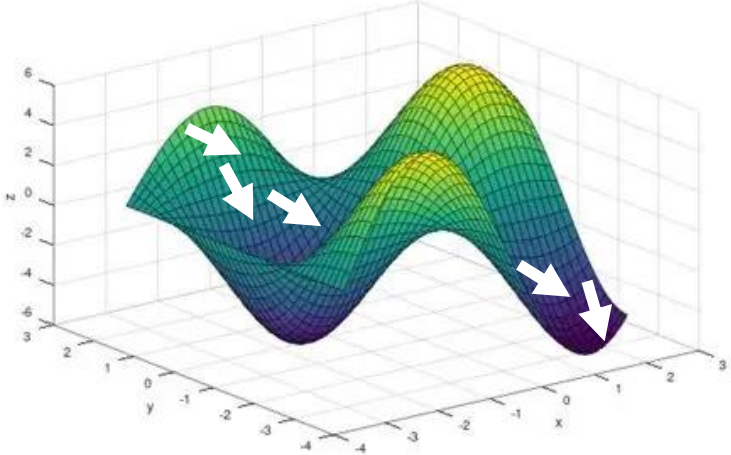


- What we want to minimize.
- In nature, environments are non-stationary, so the “objective” itself can change with context and development.
- Eg: Adolescents might prioritize finding partner/safety in group (**minimize rejection**), gonadal hormones interact with dopamine, presence of peers increase risk taking.
- Locusts change from solitary to gregarious when food becomes scarce.

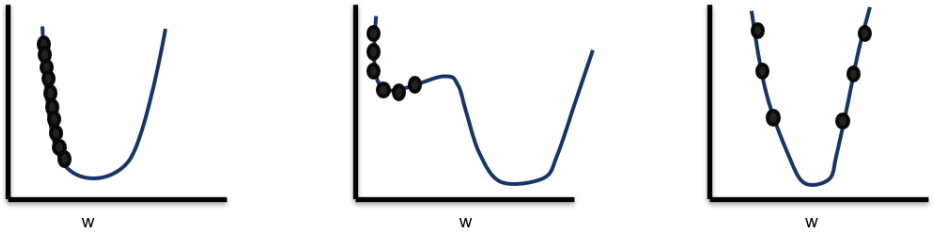
https://www.youtube.com/shorts/pNSKPLRon2o

Stochastic gradient descent

- Iterative process of adjusting connection weights to find the "minimum" of the loss function.
- Humans learn incrementally by informed trial and error
- Exploratory methods or "whims": Humans often find solutions through a sudden change of mind or curiosity.
- **Lévy walk**—many short movements in a narrow area followed by occasional long, random leaps to distant areas, avoids getting stuck in local minima.

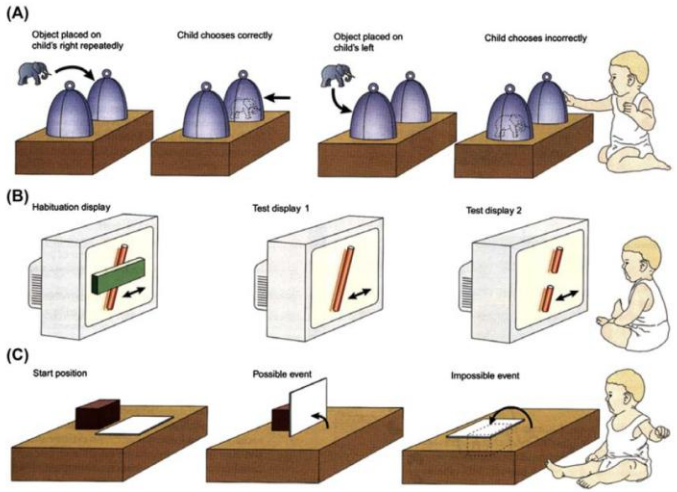


Learning rate



- Stubborn vs gullible
- We have **metacognition**: adjust learnign rate.
- **Bayesian updating**: weight that decides how much to trust the new measurement vs. the prior. When measurements are reliable or the world is changing fast, the gain (learning rate) rises; when measurements are noisy or the world is stable, it falls.
- Pupil excitement when learning: **Surprise** facilitates learning.

Posterior ← Prior + learning_rate × (surprise)



Fundamentals of Cognitive Neuroscience. Nicole Gage and Bernard Baars p 476

Surprise-based learning

Titans: Learning to Memorize at Test Time

Ali Behrouz[†], Peilin Zhong[†], and Vahab Mirrokni[†]

[†]Google Research
{alibehrouz, peilinz, mirrokni}@google.com

An event that breaks expectations (surprising) is more memorable for humans (Mandler 2014).

Inspired by this, a simple definition of surprise for a model can be its gradient with respect to the input.

$$\mathcal{M}_t = \mathcal{M}_{t-1} - \theta_t \underbrace{\nabla \ell(\mathcal{M}_{t-1}; x_t)}_{\text{Surprise}}.$$

$$\mathcal{M}_t = \mathcal{M}_{t-1} + S_t,$$
$$S_t = \eta_t \underbrace{S_{t-1}}_{\text{Past Surprise}} - \theta_t \underbrace{\nabla \ell\left(M_{t-1}; x_t\right)}_{\text{Momentary Surprise}}.$$

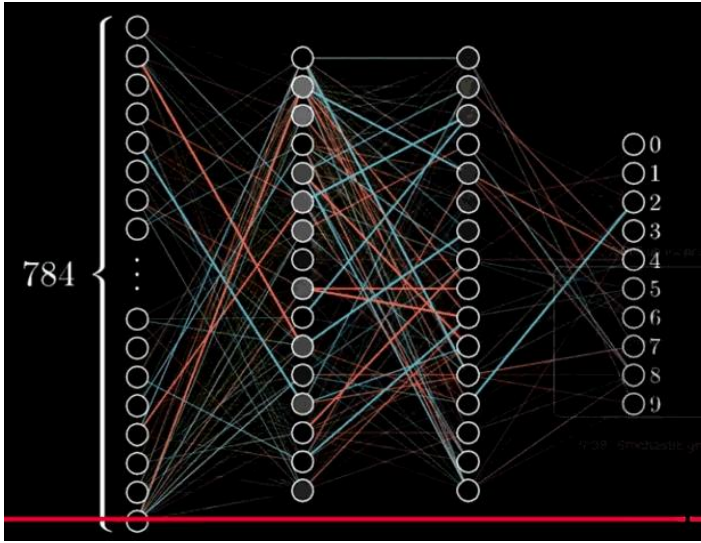
https://research.google/blog/titans-miras-helping-ai-have-long-term-memory/

Back propagation

- No evidence that biological networks can feed an error signal for modifying connection strengths any further back than the immediately preceding layer of neurons.

- We can distribute blame/credit: debug, analyze what part of the reasoning failed

- Can we implement backpropagation in brain organoids?



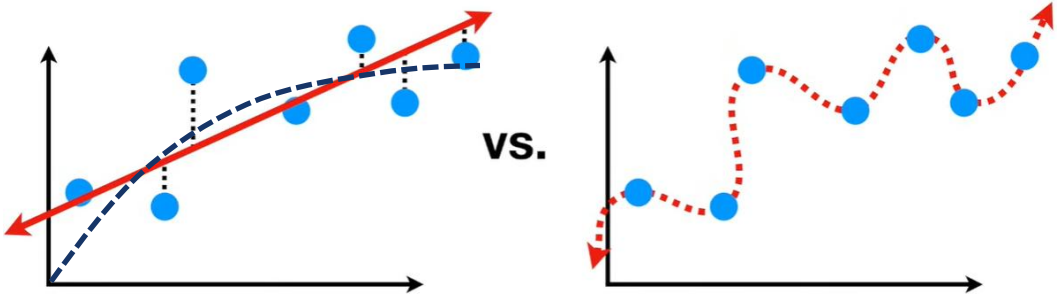
Additional ML concepts

- Bias vs Variance
- Regularization

Bias and Variance

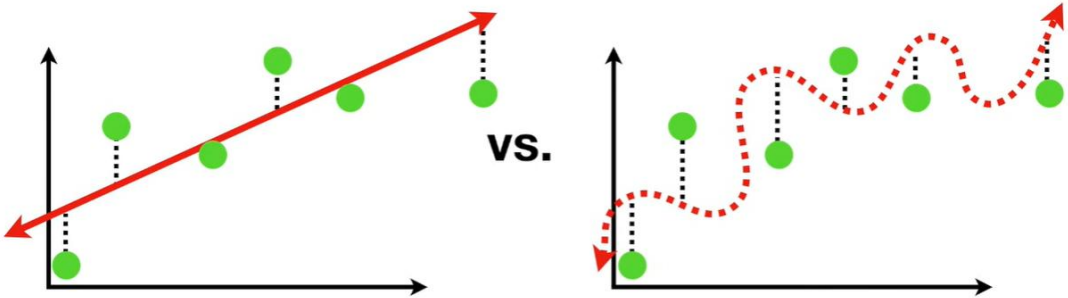
Blue: training

Squiggly line has 0 variance
straight line has bias and
some variance



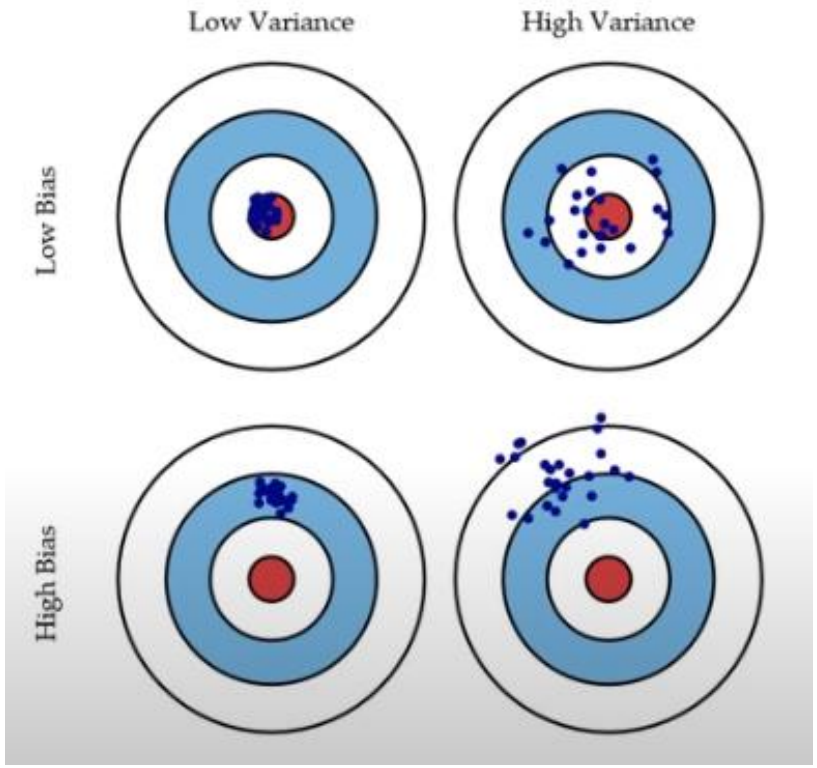
Green: test

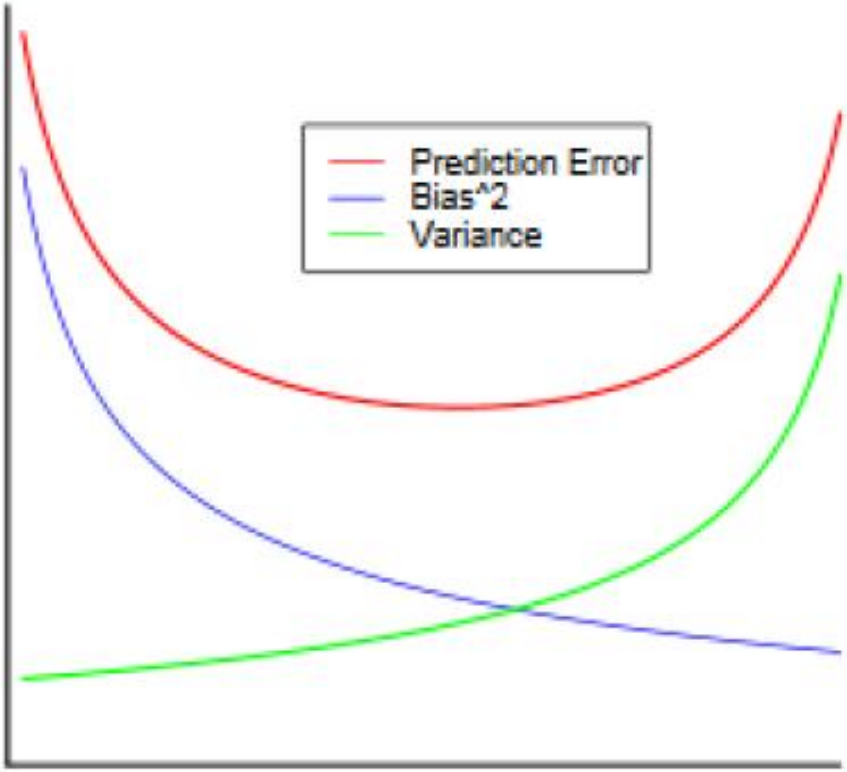
In test, the straight line has
some bias and variance, but
the squigly line has a very
high variance



StatQuest: https://www.youtube.com/watch?v=Q81RR3yKn30

Bias & Variance

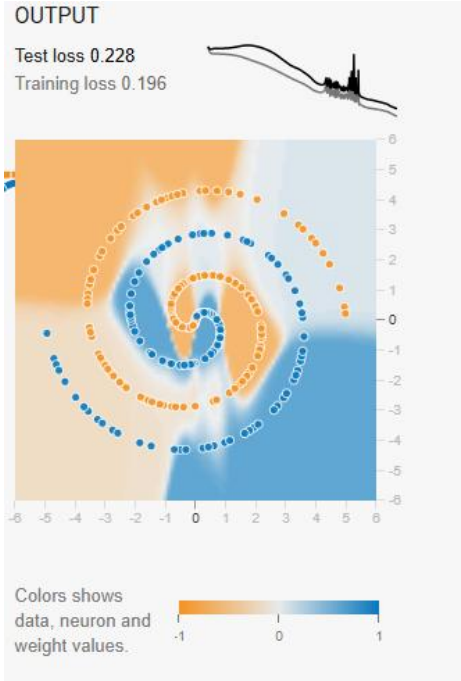




Regularization

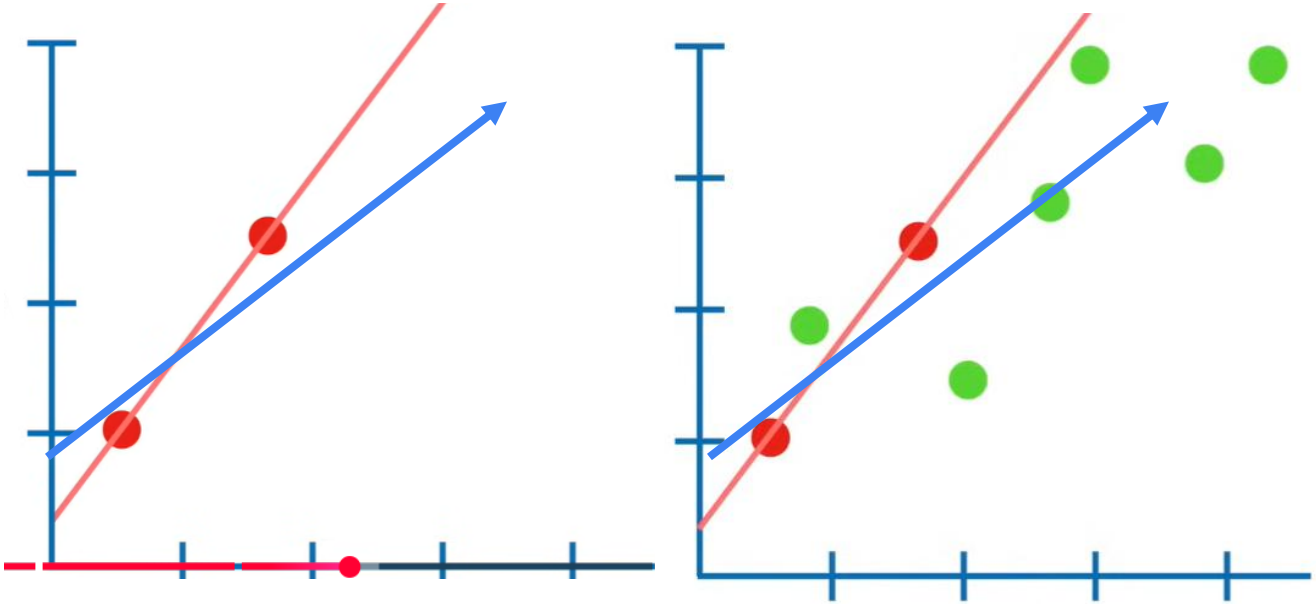
- **Prevent overfit** by impossing a penalty/limitation during training ==> reduce variance and make the predictions less sensitive to the training data

- Other regularizarion techniques:
  - Limit number of epochs (iterations)
  - Data augmentation or undersampling
  - Noise injection
  - Drop out (robustness through redundancy)



Regularization: L1 or L2

- Imposing a small penalty in training



StatQuest: https://www.youtube.com/watch?v=Q81RR3yKn30

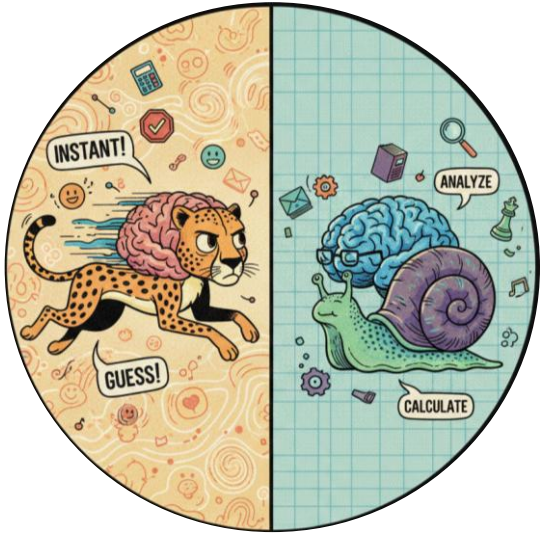
Bias and its evolutionary mechanisms

Bias is not inherently bad because it often serves a functional or adaptive purpose:

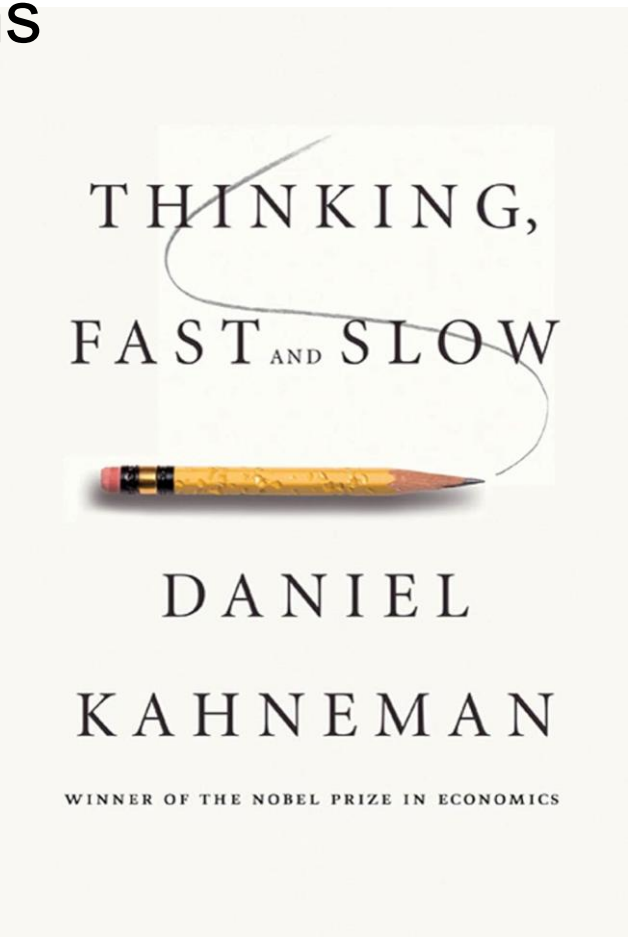
1) Survival Utility

Bias and its evolutionary mechanisms

2) Cognitive Efficiency, heuristics in a complex world.



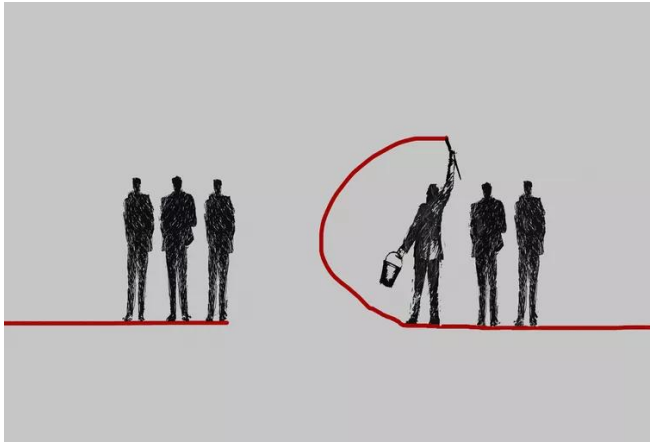
3) Predictive Power in an uncertain world: Natural events tend to have regularities



Bias and its evolutionary mechanisms

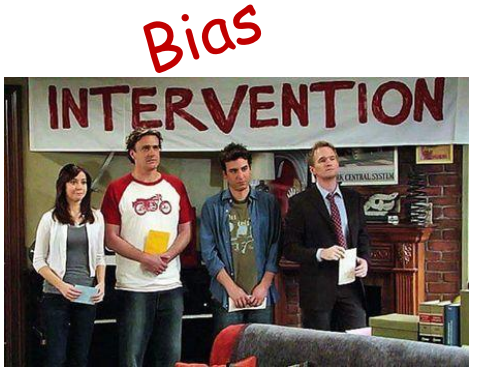
4) Attribution Tendencies: prefer internal or dispositional traits as the best explanation for others' behavior rather than considering the situation.

5) In group and outgroup bias: stronger when resources are limited



https://www.thoughtco.com/social-identity-theory-4174315

Bias and its evolutionary mechanisms

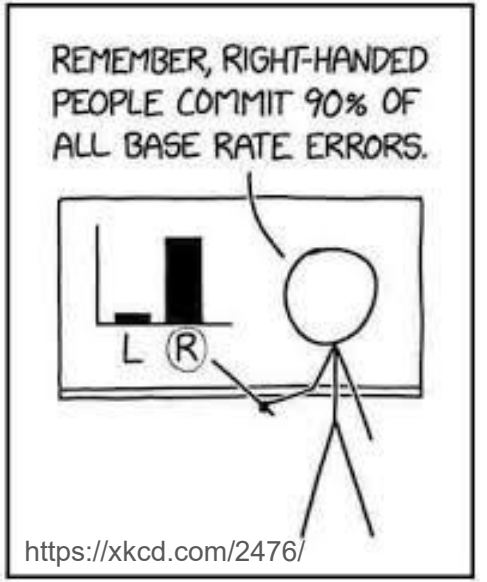


Problems:

confirmation bias

https://www.reddit.com/r/comics/comments/8jldtn/confirmation_bias_oc/?rdt=61995

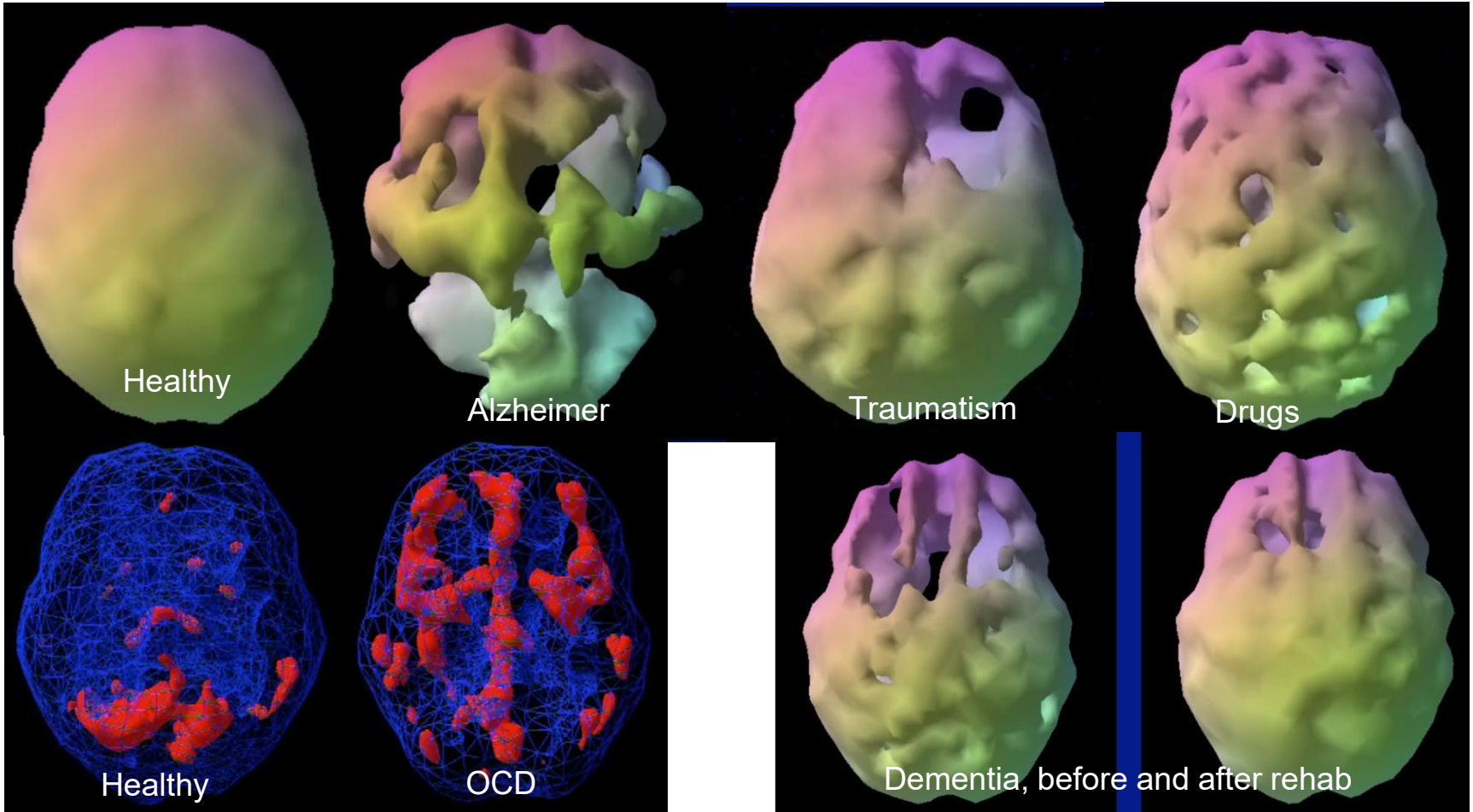
base-rate fallacy



sunk-cost fallacy

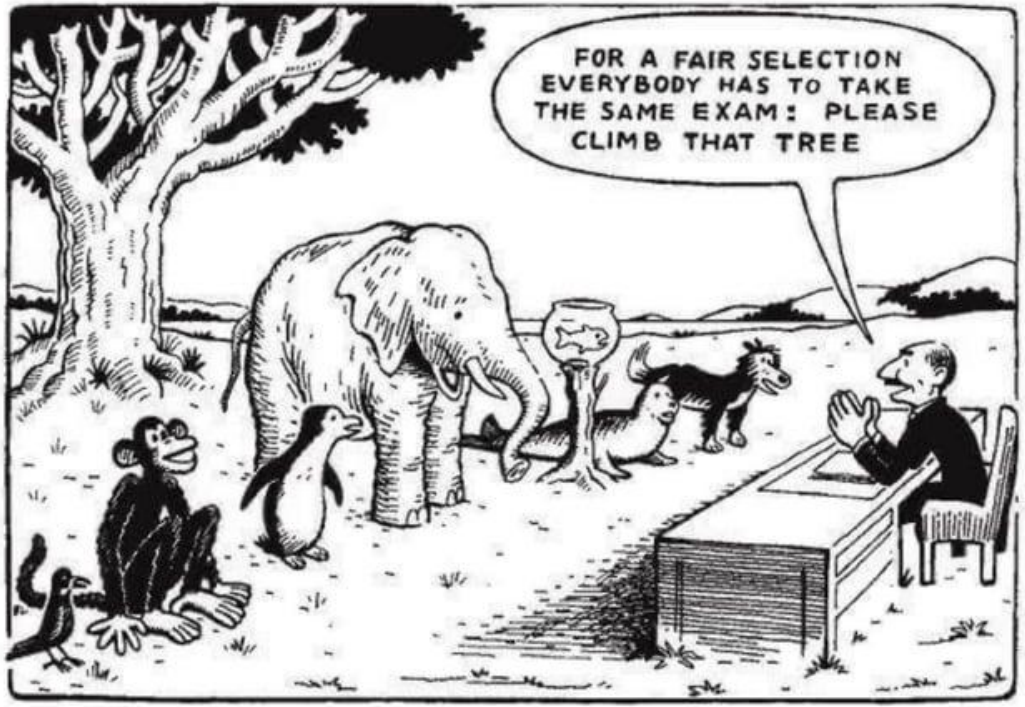
Brain plasticity

Daniel Amen TedTak: https://www.youtube.com/watch?v=esPRsT-lmw8



Brain plasticity

- True long term learning modifies your brain
  - Cab drivers → geospatial awareness
  - Braille → high hand sensitivity
  - Right/left handed people
- People learn differently → personalized approaches
- Negative effects: forgetting, additions, chronic pain,



Brain plasticity: disgust

- Innate and learnt
- Insula and Putamen regions in the brain.

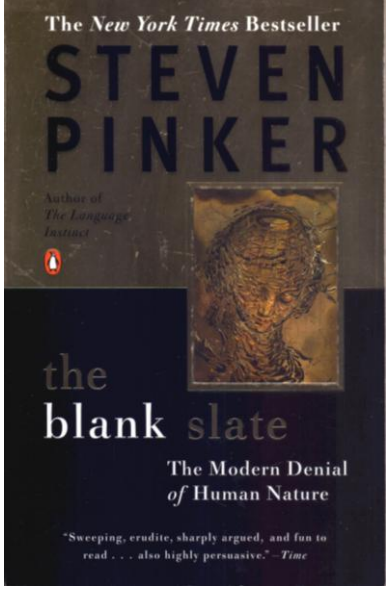
- **Core disgust** (e.g., spoiled food)
- **Animal-reminder disgust** (e.g., death, body envelope violations)
- **Contamination disgust** (e.g., hygiene cues)

Nature vs. nurture in humans

- Biological Predispositions: innate skills and preferences

- **Epigenetics**: sensory input can change **gene expression**
- Intelligence: 50% 80% genetic, but changes with age
- Wilson Effect

- Postnatal brain volume increases fourfold through **synaptogenesis and dendritic arborization**
- Environmental impact on IQ.



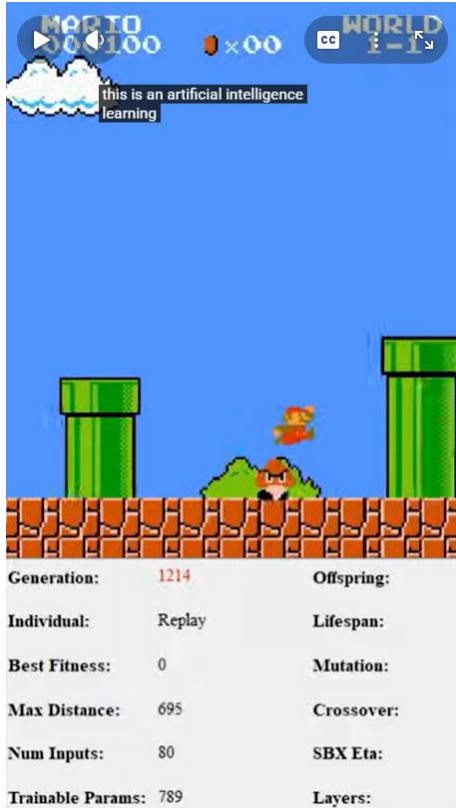
Genie, the feral child

Nature vs. nurture in AI

- Good old fashion AI: IF-THEN

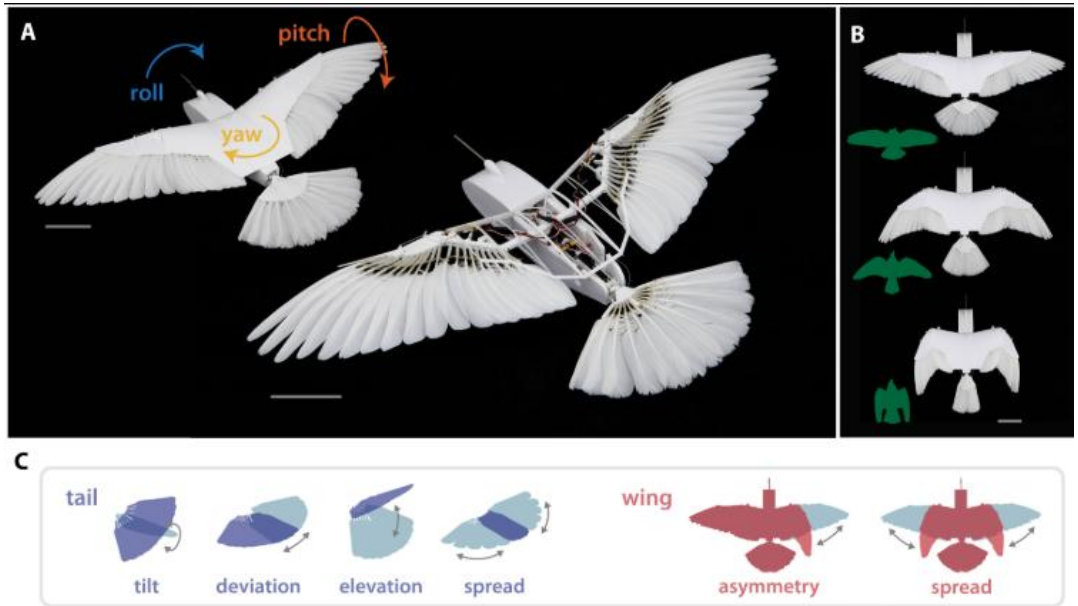
- Genetic algorithms
- Protect old knowledge while remaining plastic enough

- Weight adjustment/backprop
- GIGO → Garbage in Garbage out

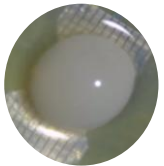


Activity: make a summary (table, mind map...)

- Planes were initially inspired by birds, but we had to find alternatives.



- ANN were also initially inspired by biological neurons, but we had to find alternatives to their mechanisms.
- Can we now mimick better the behaviour of biological neurons?



Activity: make a summary (table, mind map...)

Concept	ANN	Biological equivalent
Loss function		
Backpropagation		
Learning		
Forgetting		
Plasticity		
Regularization		
Bias		
Pattern completion		
Metacognition		
Nature vs Nurture		